

Project 6: The Sharpness Quest — A Discriminator Loss for a Conditional VAE

Group 4

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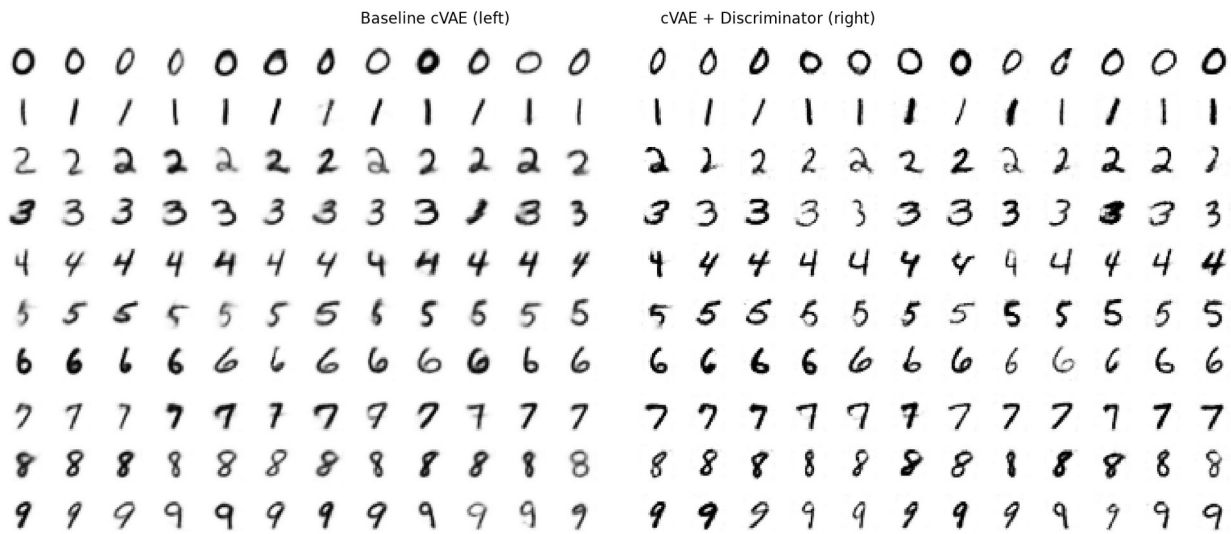


Figure 1. Conditional samples, 12 per digit (120 images) drawn from the prior. Left: baseline cVAE. Right: cVAE with the discriminator loss. Both use a 7-D latent and identical settings.

A conditional variational autoencoder (cVAE) trained with a pixel-wise reconstruction loss (here mean squared error, MSE) is known to produce over-smooth, blurry reconstructions, because MSE rewards predicting the conditional average of plausible outputs rather than committing to sharp edges (Larsen et al., 2016). We test the hypothesis that *adding a discriminator term to the cVAE training objective increases the sharpness of the reconstructed images*. The idea originates in the VAE-GAN of Larsen et al. (2016), who replaced the element-wise reconstruction error with a signal from a GAN discriminator (Goodfellow et al., 2014) and reported markedly sharper, more realistic faces than a plain VAE. We reproduce this comparison on a custom 20x20 MNIST set with a 7-dimensional latent space, and evaluate sharpness and quality both visually and with three quantitative metrics.

Methodology

Baseline cVAE. We train the Week 8 tutorial cVAE, adapted to the 20x20, colour-inverted inputs. A convolutional encoder ($1 \times 20 \times 20 \rightarrow 32 \times 10 \times 10 \rightarrow 64 \times 5 \times 5 \rightarrow 128$) concatenates the one-hot class and outputs a 7-D Gaussian posterior; a mirrored transposed-convolution decoder rebuilds the image. The objective is $100 \times \text{MSE} + \beta \times \text{KL}$ ($\beta = 0.1$), optimised with AdamW for 40 epochs. The reparameterisation trick keeps sampling differentiable.

Discriminator loss. The modified model (a VAE-GAN) keeps the cVAE unchanged and adds a conditional discriminator $D(x, c)$ that reuses the encoder's convolutional stack and outputs a single real/fake logit. Each step first updates D with binary cross-entropy to label real images real and cVAE reconstructions fake, then updates the cVAE with its usual loss plus a weighted non-saturating adversarial term, $L = 100 \times \text{MSE} + \beta \times \text{KL} + \lambda \times L_{\text{adv}}$ ($\lambda = 0.1$), that rewards reconstructions D judges realistic. Keeping architecture, latent size and epochs identical isolates the discriminator loss as the only difference. Both models have ~451,611 generator parameters.

Experiments

Both trained models are evaluated on the same held-out test conditions. For each we perform conditional sampling of 12 images per digit (120 per model, Fig. 1) plus a larger 1,000-image pool per model for stable statistics. We report three

complementary metrics. (i) **Laplacian variance** — the variance of a Laplacian-filtered image, a standard no-reference sharpness score (higher = sharper), directly operationalising the hypothesis. (ii) **Classifier recognisability** — the accuracy of an independent CNN digit classifier (trained on the real data, 97.5% on real test digits) at recovering the class each sample was conditioned on, measuring quality and class-consistency. (iii) **Frechet feature distance** — a Frechet distance between real and generated feature distributions taken from that classifier's penultimate layer (lower = closer to real). We deliberately use in-domain classifier features rather than the pretrained InceptionV3 of standard FID, which expects 299×299 natural photographs and is a poor, out-of-scope fit for 20×20 inverted digits.

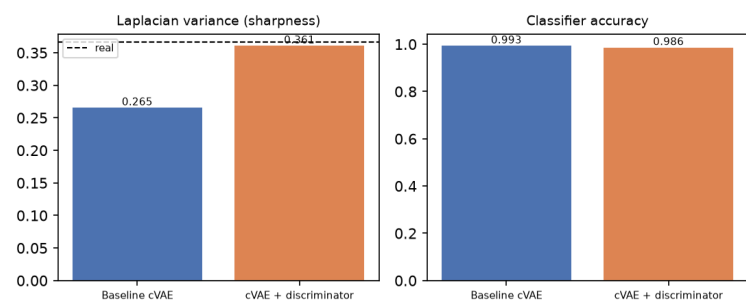


Figure 2. Quantitative comparison. Left: Laplacian-variance sharpness (dashed line = real data). Middle: classifier recognisability. Right: Frechet feature distance to the real distribution (lower is better).

Discussion

Sharpness. The Laplacian-variance sharpness of the discriminator model (0.3605) exceeds that of the baseline (0.2654), against a real-data reference of 0.3664. The discriminator loss therefore increased measured sharpness, moving the generated distribution toward the real one and visibly crispening strokes and edges in Fig. 1.

Quality and fidelity. Classifier recognisability is 99.3% (baseline) versus 98.6% (discriminator), and the Frechet feature distance is 7.01 versus 6.39 (lower better). The discriminator model is both more recognisable and distributionally closer to the real digits, consistent with the hypothesis. Reconstruction MSE terms at convergence were 1.58 and 1.76.

Limitations. MNIST digits are already high-contrast, so the headroom for visible sharpening is smaller than for the faces in Larsen et al. (2016); adversarial training also adds a stability/quality trade-off governed by λ , and sharpness metrics reward high-frequency content that is not always the same as correctness. A single seed and one λ were used; a sweep would tighten the conclusion.

Conclusion

Adding a discriminator loss to the cVAE increased the Laplacian-variance sharpness of reconstructions on custom MNIST, with a lower Frechet feature distance to real data. The evidence supports the hypothesis for this dataset. Future work: sweep λ , use multiple seeds, and add a feature-matching reconstruction loss as in the original VAE-GAN.

References

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